CHAPTER 2

H.C.F. and L.C.M. of Numbers

COMMON FACTOR

A *common factor* of two or more numbers is a number which divides each of them exactly.
For example, 4 is a common factor of 8 and 12.

HIGHEST COMMON FACTOR

Highest common factor of two or more numbers is the greatest number that divides each one of them exactly. For example, 6 is the highest common factor of 12, 18 and 24. Highest Common Factor is also called *Greatest Common Divisor* or *Greatest Common Measure*.

Symbolically, these can be written as H.C.F. or G.C.D. or G.C.M., respectively.

METHODS OF FINDING H.C.F.

I. Method of Prime Factors

Step 1 Express each one of the given numbers as the product of prime factors.
[A number is said to be a *prime number* if it is exactly divisible by 1 and itself, but not by any other number, e.g., 2, 3, 5, 7, etc. are prime numbers]

Step 2 Choose common factors.

Step 3 Find the product of these common factors. This is the required H.C.F. of given numbers.

Illustration 1: Find the H.C.F. of 70 and 90.

Solution: $70 = 2 \times 5 \times 7$

$90 = 2 \times 5 \times 9$

Common factors are 2 and 5.

$\therefore$ H.C.F. $= 2 \times 5 = 10$.

Illustration 2: Find the H.C.F. of 3332, 3724 and 4508.

Solution: $3332 = 2 \times 2 \times 7 \times 7 \times 17$

$3724 = 2 \times 2 \times 7 \times 7 \times 19$

$4508 = 2 \times 2 \times 7 \times 7 \times 23$

$\therefore$ H.C.F. $= 2 \times 2 \times 7 \times 7 = 196$.

Illustration 3: Find the H.C.F. of 360 and 132.

Solution: $360 = 2^3 \times 3^2 \times 5$

$132 = 2^2 \times 3^1 \times 11$

$\therefore$ H.C.F. $= 2^2 \times 3^1 = 12$.

Illustration 4: If $x = 2^3 \times 3^5 \times 5^9$ and $y = 2^5 \times 3^7 \times 5^{11}$, find H.C.F. of x and y.

Solution: The factors common to both x and y are 2^3, 3^5 and 5^9.

$\therefore$ H.C.F. $= 2^3 \times 3^5 \times 5^9$.

II. Method of Division

***A. For two numbers*:**

Step 1 Greater number is divided by the smaller one.

Step 2 Divisor of (1) is divided by its remainder.

Step 3 Divisor of (2) is divided by its remainder. This is continued until no remainder is left.
H.C.F. is the divisor of last step.

Illustration 5: Find the H.C.F. of 3556 and 3444.

$$
\begin{array}{r}
3444\overline{)3556}(\ 1 \\
3444 \\
\hline
112\overline{)3444}(\ 30 \\
3360 \\
\hline
84\overline{)112}(\ 1 \\
84 \\
\hline
28\overline{)\ 84}(\ 3 \\
84 \\
\hline
\times
\end{array}
$$

$\therefore$ H.C.F. $= 28$.

***B. For more than two numbers*:**

Step 1 Any two numbers are chosen and their H.C.F. is obtained.

Step 2 H.C.F. of H.C.F. (of (1)) and any other number is obtained.

Step 3 H.C.F. of H.C.F. (of (2)) and any other number (not chosen earlier) is obtained.
This process is continued until all numbers have been chosen. H.C.F. of last step is the required H.C.F.

Illustration 6: Find the H.C.F. of 13915, 9499 and 2553 by division method.

Solution:

$$
\begin{array}{r|l}
9499 & 13915 \\
 & 9499 \\
\hline
4416 & 9499 \\
 & 8832 \\
\hline
667 & 4416 \\
 & 4002 \\
\hline
414 & 667 \\
 & 414 \\
\hline
253 & 414 \\
 & 253 \\
\hline
161 & 253 \\
 & 161 \\
\hline
92 & 161 \\
 & 92 \\
\hline
69 & 92 \\
 & 69 \\
\hline
69 & 69 \\
 & 69 \\
\hline
 & \times
\end{array}
\quad
\begin{array}{l}
1 \\
 \\
2 \\
 \\
6 \\
 \\
1 \\
 \\
1 \\
 \\
1 \\
 \\
1 \\
 \\
3 \\
 \\
 \\

Illustration 9: Find the L.C.M. of 12, 15, 20 and 54.

Solution:

2	12,	15,	20,	54
2	6,	15,	10,	27
3	3,	15,	5,	27
5	1,	5,	5,	9
	1,	1,	1,	9

L.C.M. = 2 × 2 × 3 × 5 × 1 × 1 × 1 × 9
= 540.

Notes

Before finding the L.C.M. or H.C.F., we must ensure that all quantities are expressed in the same unit.

SHORT-CUT METHODS

01 H.C.F. and L.C.M. of Decimals

Step 1 Make the same number of decimal places in all the given numbers by suffixing zero(s) if necessary.

Step 2 Find the H.C.F./L.C.M. of these numbers without decimal.

Step 3 Put the decimal point (in the H.C.F./L.C.M. of Step 2) leaving as many digits on its right as there are in each of the numbers.

Illustration 10: Find the L.C.M. of 1.2, 0.24 and 6.

Solution: The given numbers can be written as 1.20, 0.24 and 6.00.

Now, ignoring the decimal we find the L.C.M. of 120, 24 and 600.

2	120,	24,	600
2	60,	12,	300
2	30,	6,	150
3	15,	3,	75
5	5,	1,	25
	1,	1,	5

∴ L.C.M. = 2 × 2 × 2 × 3 × 5 × 1 × 5 = 600

Thus, the required L.C.M. = 6.00, i.e., 6.

Illustration 11: Find the H.C.F. of 6.16 and 13.

Solution: The given numbers can be written as 6.16 and 13.00.

Now, ignoring the decimals we find the H.C.F. of 616 and 1300.

$$\begin{array}{r|l} 616 & 1300 \\ & 1232 \\ \hline 68 & 616 \\ & 612 \\ \hline 4 & 68 \\ & 68 \\ \hline & \times \end{array} \quad \begin{array}{l} \big) \\ \big(2 \\ \\ \big) \\ \big(9 \\ \\ \big) \\ \big(17 \\ \\ \end{array}$$

∴ H.C.F. of 616 and 1300 is 4.

Thus, the required H.C.F. = 0.04.

02 L.C.M. and H.C.F. of Fractions

$$\text{L.C.M} = \frac{\text{L.C.M. of the numbers in numerators}}{\text{H.C.F. of the numbers in denominators}}$$

$$\text{H.C.F.} = \frac{\text{H.C.F. of the numbers in numerators}}{\text{L.C.M. of the numbers in denominators}}$$

Illustration 12: Find the L.C.M. of $\frac{2}{5}$, $\frac{3}{10}$ and $\frac{6}{25}$.

Solution: L.C.M. of numerators 2, 3 and 6 is 6.

H.C.F. of denominators 5, 10 and 25 is 5.

∴ Required L.C.M. $= \frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$

$= \frac{6}{5}$.

Illustration 13: Find the H.C.F. of $\frac{4}{9}$, $\frac{10}{21}$ and $\frac{20}{63}$.

Solution: H.C.F. of numerators 4, 10 and 20 is 2.

L.C.M. of denominators 9, 21 and 63 is 63.

∴ Required H.C.F $= \frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}} = \frac{2}{63}$.

Notes

1. If the given set of numbers includes fractions as well as whole numbers, treat whole number too as fraction with 1 in its denominator.
2. The H.C.F. of a number of fractions is always a fraction, but the L.C.M. may be a fraction or an integer.

03 Product of two numbers
= L.C.M. of the numbers × H.C.F. of the numbers

Illustration 14: The H.C.F. and the L.C.M. of any two numbers are 63 and 1260, respectively. If one of the two numbers is 315, find the other number.

Solution: The required number

$$= \frac{\text{L.C.M.} \times \text{H.C.F.}}{\text{First Number}} = \frac{1260 \times 63}{315} = 252.$$

04 To find the greatest number that will exactly divide x, y and z.
Required number = H.C.F. of x, y and z.

Illustration 15: Find the greatest number that will exactly divide 200 and 320.

Solution: The required greatest number
= H.C.F. of 200 and 320 = 40.

05 To find the greatest number that will divide x, y and z leaving remainders a, b and c, respectively.
Required number = H.C.F. of $(x - a)$, $(y - b)$ and $(z - c)$.

Illustration 16: Find the greatest number that will divide 148, 246 and 623 leaving remainders 4, 6 and 11, respectively.

Solution: The required greatest number
= H.C.F. of (148 – 4), (246 – 6) and (623 – 11),
i.e., H.C.F. of 144, 240 and 612 = 12.

06 To find the least number which is exactly divisible by x, y and z.
Required number = L.C.M. of x, y and z.

Illustration 17: What is the smallest number which is exactly divisible by 36, 45, 63 and 80?

Solution: The required smallest number
= L.C.M. of 36, 45, 63 and 80
= 5040.

07 To find the least number which when divided by x, y and z leaves the remainders a, b and c, respectively. It is always observed that $(x - a) = (y - b) = (z - c) = k$(say)
$\therefore$ Required number = (L.C.M. of x, y and z) $- k$.

Illustration 18: Find the least number which when divided by 36, 48 and 64 leaves the remainders 25, 37 and 53, respectively.

Solution: Since, (36 – 25) = (48 – 37) = (64 – 53) = 11, therefore, the required smallest number
= (L.C.M. of 36, 48 and 64) – 11
= 576 – 11 = 565.

08 To find the least number which when divided by x, y and z leaves the same remainder r in each case.
Required number = (L.C.M. of x, y and z) + r.

Illustration 19: Find the least number which when divided by 12, 16 and 18, will leave in each case a remainder 5.

Solution: The required smallest number
= (L.C.M. of 12, 16 and 18) + 5
= 144 + 5 = 149.

09 To find the greatest number that will divide x, y and z leaving the same remainder in each case.

(a) When the value of remainder r is given:
Required number = H.C.F. of $(x - r)$, $(y - r)$ and $(z - r)$.

(b) When the value of remainder is not given:
Required number = H.C.F. of $|(x - y)|$, $|(y - z)|$ and $|(z - x)|$

Illustration 20: Find the greatest number which will divide 772 and 2778 so as to leave the remainder 5 in each case.

Solution: The required greatest number
= H.C.F. of (772 – 5) and (2778 – 5)
= H.C.F. of 767 and 2773
= 59.

Illustration 21: Find the greatest number which on dividing 152, 277 and 427 leaves equal remainder.

Solution: The required greatest number
= H.C.F. of $|(x - y)|$, $|(y - z)|$ and $|(z - x)|$
= H.C.F. of $|(152 - 277)|$, $|(277 - 427)|$ and $|(427 - 152)|$
= H.C.F. of 125, 150 and 275
= 25.

10 To find the n-digit greatest number which, when divided by x, y and z,

(a) leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and $z = L$

Step 2 $L\overline{)n\text{-digit greatest number}}$
Remainder = R

Step 3 Required number
= n-digit greatest number $- R$

(b) leaves remainder K in each case
Required number
= (n-digit greatest number $- R$) + K.

Illustration 22: Find the greatest number of 4-digit number which, when divided by 12, 18, 21 and 28 leaves 3 as a remainder in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252.

$$252\overline{)9999}\,(39$$
$$\underline{9828}$$
$$171$$

$\therefore$ The required number = (9999 – 171) + 3 = 9931.

Illustration 23: Find the greatest number of four digits which, when divided by 12, 15, 20 and 35 leaves no remainder.

Solution: L.C.M. of 12, 15, 20 and 35 = 420.

$$420\overline{)9999}\,(23$$
$$\underline{9660}$$
$$339$$

$\therefore$ The required number = 9999 – 339
= 9663.

11 To find the n-digit smallest number which when divided by x, y and z

(a) leaves no remainder (i.e., exactly divisible)

Step 1 L.C.M. of x, y and $z = L$

Step 2 $L\overline{)n\text{-digit smallest number}\,(}$
Remainder = R

Step 3 Required number
= n-digit smallest number + $(L - R)$.

(b) leaves remainder K in each case.
Required number
= n-digit smallest number + $(L - R) + K$.

Illustration 24: Find the least number of four digits which is divisible by 4, 6, 8 and 10.

Solution: L.C.M. of 4, 6, 8 and 10 = 120.

$$120\overline{)1000}(8$$
$$\underline{960}$$
$$40$$

$\therefore$ The required number = 1000 + (120 – 40)
= 1080.

Illustration 25: Find the smallest 4-digit number, such that when divided by 12, 18, 21 and 28, it leaves remainder 3 in each case.

Solution: L.C.M. of 12, 18, 21 and 28 = 252.

$$252\overline{)1000}\,(3$$
$$\underline{756}$$
$$244$$

$\therefore$ The required number
= 1000 + (252 – 244) + 3
= 1011.

EXERCISE-I

1. What is the H.C.F. of 27, 18 and 36?

(a) 7 (b) 11
(c) 9 (d) None of these

2. Determine the L.C.M of $\frac{2}{5}, \frac{3}{10}$ and $\frac{6}{25}$.

(a) $\frac{6}{5}$ (b) $\frac{11}{5}$
(c) $\frac{9}{5}$ (d) None of these

3. What is the L.C.M. of 25, 30, 35 and 40?

(a) 3800
(b) 4200
(c) 4400
(d) None of these

4. What is the greatest number which divides 852, 1065 and 1491 exactly?

(a) 193 (b) 183
(c) 223 (d) 213

5. What is the H.C.F. of $\frac{4}{9}, \frac{10}{21}$ and $\frac{20}{30}$?

(a) $\frac{4}{189}$ (b) $\frac{6}{23}$
(c) $\frac{2}{63}$ (d) None of these

6. Find the least number which when divided by 16, 18, 20 and 25 leaves 4 as remainder in each case but when divided by 7 leaves no remainder.

(a) 8004 (b) 13004
(c) 18004 (d) 18014

7. Area of three fields is 165 m^2, 195 m^2 and 85 m^2, respectively. In each of the fields a flower bed of equal length has to be made. If flower bed in each

of the fields is 3 m wide then what is the maximum length of the flower bed in each of the fields?

(a) 7 m (b) 9 m
(c) 5 m (d) None of these

8. Find the greatest number which will divide 2112 and 2792 leaving the remainder 4 in each case.

(a) 78 (b) 68
(c) 65 (d) 63

9. The H.C.F. of two numbers is 12 and their difference is 12. The numbers are:

(a) 66, 78 (b) 70, 82
(c) 94, 106 (d) 84, 96

10. A merchant has 435 litres, 493 litres and 551 litres of three different kinds of milk. Find the least number of casks of equal size required to store all the milk without mixing.

(a) 51 (b) 61
(c) 47 (d) 45

11. Find the greatest number which will divide 25, 73 and 97 so as to leave the same remainder in each case.

(a) 12 (b) 18
(c) 24 (d) 32

12. The sum of two numbers is 216 and their H.C.F. is 27. The numbers are:

(a) 54, 162 (b) 108, 118
(c) 27, 189 (d) None of these

13. How often will five bells toll together in one hour if they start together and toll at intervals of 5, 6, 8, 12, 20 seconds, respectively?

(a) 29 (b) 30
(c) 31 (d) 120

14. Find the greatest number that will divide 964, 1238 and 1400 leaving remainders 41, 31 and 51, respectively.

(a) 71 (b) 81
(c) 61 (d) 73

15. Find the side of the largest square slabs which can be paved on the floor of a room 5 m 44 cm long and 3 m 74 cm broad.

(a) 56 (b) 42
(c) 38 (d) 34

16. The traffic lights at three different road crossings change after every 48 seconds, 72 seconds and 108 seconds, respectively. If they all change simultaneously at 8:20:00 hours; then they will again change simultaneously at:

(a) 8:27:12 hours (b) 8:27:24 hours
(c) 8:27:36 hours (d) 8:27:48 hours

17. The product of two numbers is 6760 and their H.C.F. is 13. How many such pairs can be formed?

(a) 2 (b) 3
(c) 4 (d) only one

18. Find the greatest number of four digits which when divided by 10, 15, 21 and 28 leaves 4, 9, 15 and 22 as remainders, respectively.

(a) 9654 (b) 9666
(c) 9664 (d) 9864

19. The number of prime factors in the expression $(6)^{10} \times (7)^{17} \times (11)^{27}$ is:

(a) 54 (b) 64
(c) 71 (d) 81

20. Find the greatest number which will divide 3962, 4085 and 4167 leaving the same remainder in each case.

(a) 37 (b) 39
(c) 41 (d) 43

21. A wholesale tea dealer has 408 kilograms, 468 kilograms and 516 kilograms of three different qualities of tea. He wants it all to be packed into boxes of equal size without mixing. Find the capacity of the largest possible box.

(a) 50 (b) 36
(c) 24 (d) 12

22. A room is 4 m 37 cm long and 3 m 23 cm broad. It is required to pave the floor with minimum square slabs. Find the number of slabs required for this purpose.

(a) 485 (b) 431
(c) 391 (d) 381

23. The least perfect square number which is divisible by 3, 4, 5, 6 and 8:

(a) 900 (b) 1200
(c) 2500 (d) 3600

24. Find the least number of five digits which when divided by 12, 16, 21, 36 and 40 leaves remainder 8 in each case.

(a) 10088 (b) 10072
(c) 10080 (d) None of these

25. Three pieces of timber 42 m, 49 m and 63 m long have to be divided into planks of the same length. What is the greatest possible length of each plank?

(a) 7 m (b) 14 m
(c) 42 m (d) 63 m

26. Three men start together to travel the same way around a circular track of 11 kilometres in circumference. Their speeds are 4, $5\frac{1}{2}$ and 8 Km/h, respectively. When will they meet at the starting point?

(a) 11 hours (b) 12 hours
(c) 23 hours (d) 22 hours

27. Five bells begin to toll together and toll at intervals of 36, 45, 72, 81 and 108 seconds. After what interval of time will they keep on tolling together?

(a) 3240 seconds (b) 3080 seconds
(c) 3140 seconds (d) 3200 seconds

28. Three different containers contain different quantities of mixture of milk and water, whose measurements are 403 Kg, 434 Kg and 465 Kg. What biggest measure must be there to measure all the different quantities exactly?

(a) 1 Kg (b) 7 Kg
(c) 31 Kg (d) 41 Kg

29. The L.C.M. and G.C.D. of two numbers are 1530 and 51, respectively. Find how many such pairs are possible?

(a) 2 (b) 3
(c) 4 (d) Only one

30. Find the least number of five digits which when divided by 63, 56 and 42 leaves remainder 1 in each case.

(a) 10082 (b) 10081
(c) 10001 (d) 10071

31. The H.C.F. and L.C.M. of two numbers are 44 and 264, respectively. If the first number is divided by 2, the quotient is 44. The other number is:

(a) 33 (b) 66
(c) 132 (d) 264

32. The largest natural number which exactly divides the product of any four consecutive natural numbers, is:

(a) 6 (b) 12
(c) 24 (d) 120

33. Find the least number of six digits which is exactly divisible by 15, 21 and 28:

(a) 100480 (b) 100270
(c) 100380 (d) 100340

34. Find the greatest number of five digits which when divided by 12, 15, 21, 25 and 28 leaves 5, 8, 14, 18 and 21 as remainders, respectively.

(a) 98696 (b) 98700
(c) 97693 (d) 98693

35. What is the smallest number which when increased by 3 is divisible by 16, 24, 30 and 32?

(a) 480 (b) 475
(c) 472 (d) 477

36. The least number of square tiles required to cover the ceiling of a room 15 m 17 cm long and 9 m 2 cm broad, is:

(a) 656 (b) 738
(c) 814 (d) 902

37. Find the least number which when divided by 2, 3, 4, 5 and 6 leaves 1, 2, 3, 4 and 5 as remainders, respectively, but when divided by 7 leaves no remainder.

(a) 210 (b) 119
(c) 126 (d) 154

38. Find the greatest number of five digits which when divided by 4, 6, 10 and 15 leaves the same remainder 3 in each case.

(a) 99993 (b) 99063
(c) 90093 (d) 99963

39. Find the least number which is a multiple of 31 and when divided by 15, 24 and 32 leaves the remainders 2, 11 and 19, respectively.

(a) 2418 (b) 2387
(c) 2356 (d) 2325

40. Find the two largest numbers of four digits having 531 as their H.C.F.

(a) 9231, 9762
(b) 9027, 9558
(c) 9037, 9568
(d) 9127, 9658

41. Find the greatest number of five digits which becomes exactly divisible by 10, 12, 15 and 18 when 3769 is added to it.

(a) 99811 (b) 99911
(c) 98911 (d) 99011

42. Find the least number which when decreased by 11 is divisible by 14, 15, 21, 32 and 60.

(a) 4371 (b) 3271
(c) 3371 (d) 3360

43. Find the least number of five digits which when divided by 8, 12, 16 and 20 leaves remainders 1, 5, 9 and 13, respectively.

(a) 10003 (b) 10093
(c) 10073 (d) 10013

44. The H.C.F. of two numbers is 11 and their L.C.M. is 693. If one of the numbers is 77, find the other.

(a) 909 (b) 119
(c) 66 (d) 99

45. Find the greatest number of four digits which is exactly divisible by 24, 28, 30 and 35.

(a) 9225 (b) 9240
(c) 9250 (d) 9260

46. Find the greatest number of four digits which must be added to 5231 so that the final number becomes exactly divisible by 12, 15, 27, 32 and 40.

(a) 7929 (b) 7829
(c) 9729 (d) 7729

47. A heap of stones can be made up into groups of 21. When made up into groups of 16, 20, 25 and 45, there are 3 stones left in each case. How many stones at least can there be in the heap?

(a) 7203 (b) 2403
(c) 3603 (d) 4803

48. Find the greatest number of five digits which when divided by 8, 9 and 10 leaves 3 as remainder in each case.

(a) 99996
(b) 99723
(c) 99983
(d) None of these

49. What is the least number of cut pieces of equal length that can be cut out of two lengths 10 m 857 mm and 15 m 87 mm?

(a) 174 (b) 172
(c) 164 (d) 184

EXERCISE-2
(BASED ON MEMORY)

1. *M* is the largest 4 digit number, which when divided by 4, 5, 6 and 7 leaves remainder as 2, 3, 4 and 5 respectively. When M is divided by 9?

(a) 2 (b) 1
(c) 3 (d) 6

[SSC CGL Tier-II CBE, 2018]

2. Three electronics devices make a beep after every 48 seconds, 72 seconds and 108 seconds respectively. They beeped together at 10 a.m. The time when they will next make a beep together at the earliest is

(a) 10 : 07 : 12 hours
(b) 10 : 07 : 24 hours
(c) 10 : 07 : 36 hours
(d) 10 : 07 : 48 hours

[SSC CGL Tier-II (CBE), 2017]

3. Two pipes of length 1.5 m and 1.2 m are to be cut into equal pieces without leaving any extra length of pipes. The greatest length of the pipe pieces of same size which can be cut from these two lengths will be

(a) 0.13 metre (b) 0.4 metre
(c) 0.3 metre (d) 0.41 metre

[SSC CGL Tier-II (CBE), 2017]

4. What least value which should be added to 1812 to make it divisible by 7, 11 and 14?

(a) 12 (b) 36
(c) 72 (d) 154

[SSC CHSL Tier-I CBE, 2017]

5. How many numbers are there from 300 to 700 which are divisible by 2, 3 and 7?

(a) 7 (b) 8
(c) 9 (d) 10

[SSC CGL Tier-I CBE, 2017]

6. The LCM of two prime numbers x and y. ($x > y$) is 161. The value of ($3y - x$)

(a) –2 (b) –1
(c) 1 (d) 2

[SSC CGL Tier-I (CBE), 2016]

7. The product of two 2 digit numbers is 2160 and their H.C.F. is 12. The numbers are

(a) (12, 60) (b) (72, 30)
(c) (36, 60) (d) (60, 72)

[SSC CGL Tier-I (CBE), 2016]

8. Find the greatest number that will divide 390, 495 and 300 without leaving a remainder.

(a) 5 (b) 15
(c) 25 (d) 35

[SSC CGL Tier-I (CBE), 2016]

9. In a school 391 boys and 323 girls have been divided into the largest possible equal classes, so that each class of boys numbers the same as each class of girls. What is the number of classes?

(a) 23 (b) 19
(c) 44 (d) 17

[SSC CGL Tier-I (CBE), 2016]

10. The HCF of $x^8 - 1$ and $x^4 + 2x^3 - 2x - 1$ is:

(a) $x^2 + 1$ (b) $x + 1$
(c) $x^2 - 1$ (d) $x - 1$

[SSC, 2015]

11. Three number s are in the ratio 1 : 2 : 3 and their HCF is 12. The numbers are

(a) 12, 24, 36 (b) 5, 10, 15
(c) 4, 8, 12 (d) 10, 20, 30

[SSC, 2014]

12. Find the least number which when divided separately by 15, 20, 36 and 48 leaves 3 as remainder in each case.

(a) 183 (b) 243
(c) 483 (d) 723

[SSC, 2014]

13. If the L.C.M. and H.C.F. of two expressions are $(x^2+6x+8)(x+1)$ and $(x+1)$, respectively and one of the expressions is x^2+3x+2, find the other.

(a) x^2+5x+4 (b) x^2-5x+4
(c) x^2+4x+5 (d) x^2-4x+5

[SSC, 2014]

14. What is the smallest number by which 625 must be divided so that the quotient is a perfect cube?

(a) 125 (b) 5
(c) 2 (d) 3

[SSC, 2014]

15. Find the greatest number which exactly divides 200 and 320.

(a) 10 (b) 20
(c) 16 (d) 40

[SSC, 2014]

16. The greatest 4-digit number exactly divisible by 10, 15 and 20 is:

(a) 9990 (b) 9960
(c) 9980 (d) 9995

[SSC, 2013]

17. If the students of 9th class are arranged in rows of 6, 8, 12 or 16, no student is left behind. The possible number of students in the class is:

(a) 60 (b) 72
(c) 80 (d) 96

[SSC, 2013]

18. If A and B are the H.C.F. and L.C.M., respectively of two algebraic expressions x and y, and $A + B = x + y$, then the value of $A^3 + B^3$ is:

(a) $x^3 - y^3$
(b) x^3
(c) y^3
(d) $x^3 + y^3$

[SSC Assistant Grade III, 2013]

19. The greatest number that divides 411, 684, 821 and leaves 3, 4 and 5 as remainders, respectively is:

(a) 254 (b) 146
(c) 136 (d) 204

[SSC Assistant Grade III, 2013]

20. L.C.M. and H.C.F. of two numbers x and y are 3 and 105, respectively. If $x + y = 36$, the value of $\frac{1}{x}+\frac{1}{y}$ is:

(a) 1 (b) $\frac{1}{6}$
(c) $\frac{12}{315}$ (d) $\frac{4}{35}$

[UPPCS, 2012]

21. Given: $\sqrt[3]{4}$, $\sqrt{3}$, $\sqrt[6]{25}$ and $\sqrt[12]{289}$, the greatest and least of them are respectively:

(a) $\sqrt[12]{289}$ and $\sqrt[3]{4}$
(b) $\sqrt{3}$ and $\sqrt[3]{4}$
(c) $\sqrt[6]{25}$ and $\sqrt{3}$
(d) $\sqrt[3]{4}$ and $\sqrt[6]{25}$

[SSC Assistant Grade III, 2012]

22. In four consecutive prime numbers that are in ascending order, the product of the first three is 385 and that of the last three is 1001. The largest given prime number is: